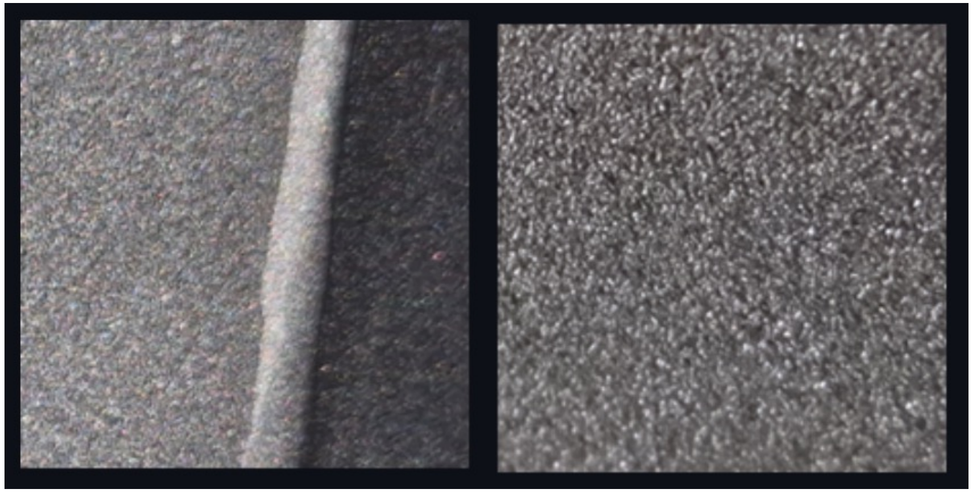
ML Models Comparison

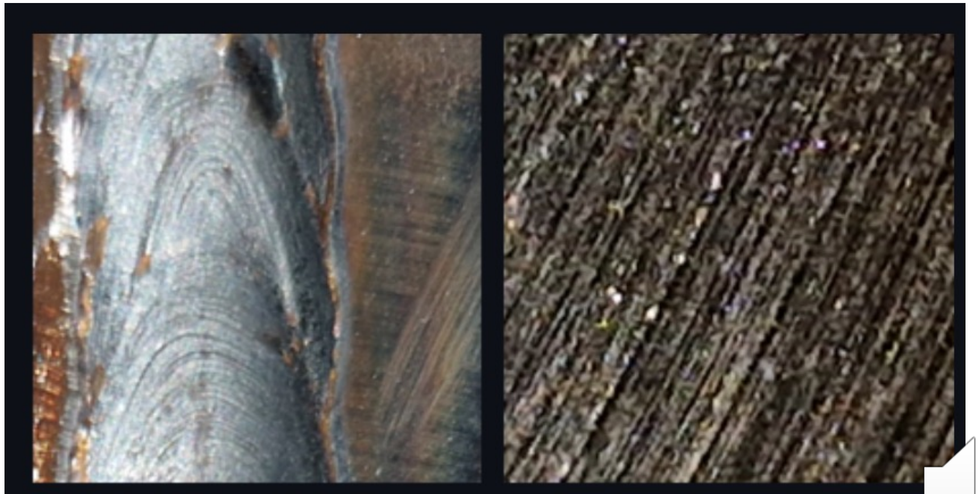
Deep Learning vs Classical ML

Image classification

- SteelBlastQC Dataset (Ruzavina *et al.*, 2025)
- Maastricht University in Netherlands

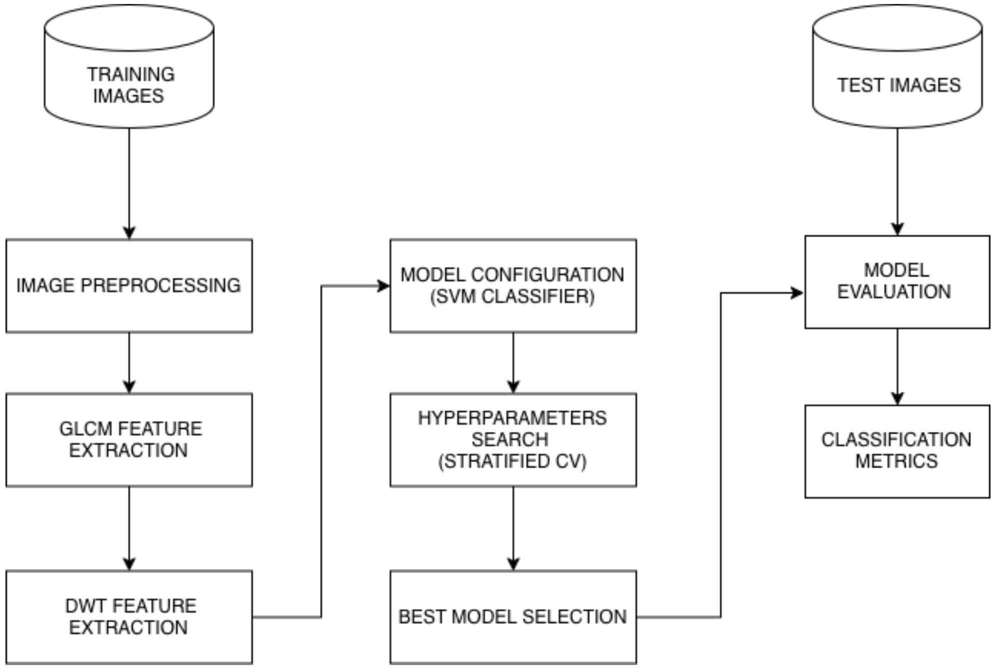


Ready for paint (negative class)

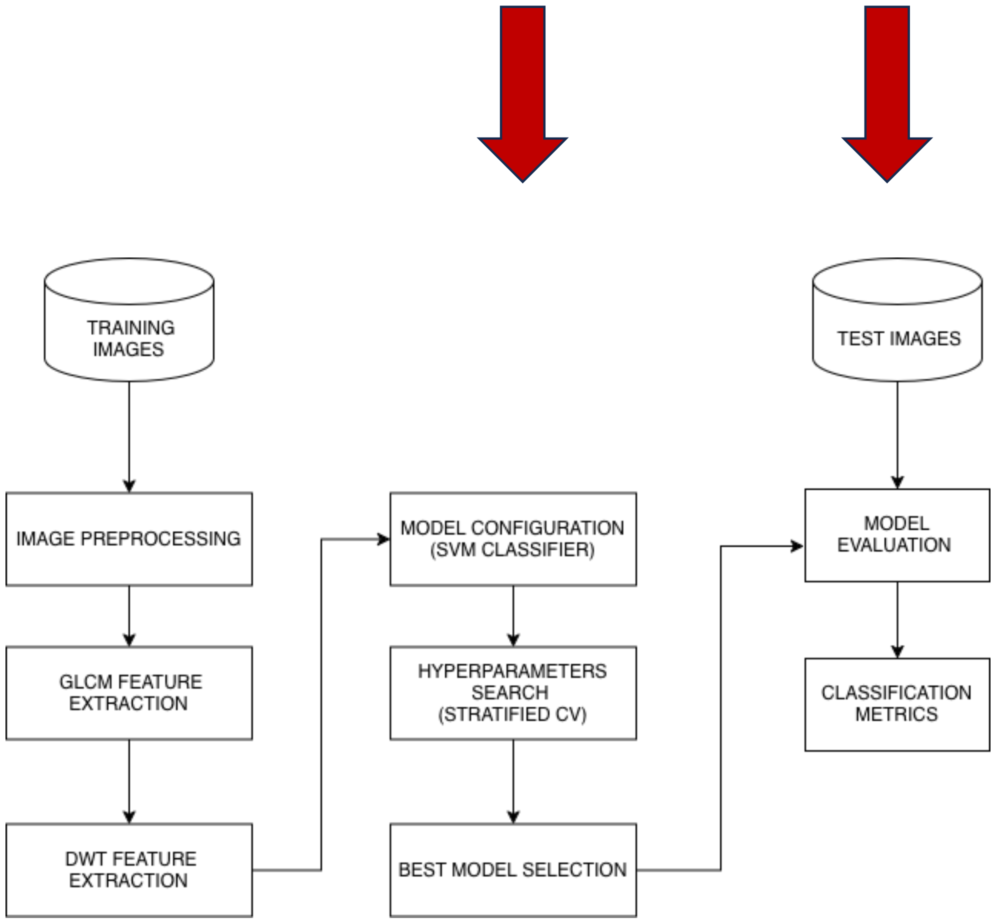


Needs shot-blasting (positive class)

Classical ML: SVM (GLCM+DWT)



- Scikit-learn
  - De facto standard for classic ML in Python
- Resize images to 256x256
- Grayscale images to emphasize texture and structure
- GLCM (Gray-Level Co-occurrence Matrix)
  - Texture feature extraction method that describes the relationship between neighboring pixels.
  - Features:
    - Contrast, Dissimilarity, Homogeneity, etc
- DWT (Discrete Wavelet Transform)
  - It decomposes an image into multiple frequency components.
  - Features:
    - Energy, Entropy, etc
- Generated features are combined to improve accuracy on tribological surfaces (Stachowiak, Podsiadlo and Stachowiak, 2005)



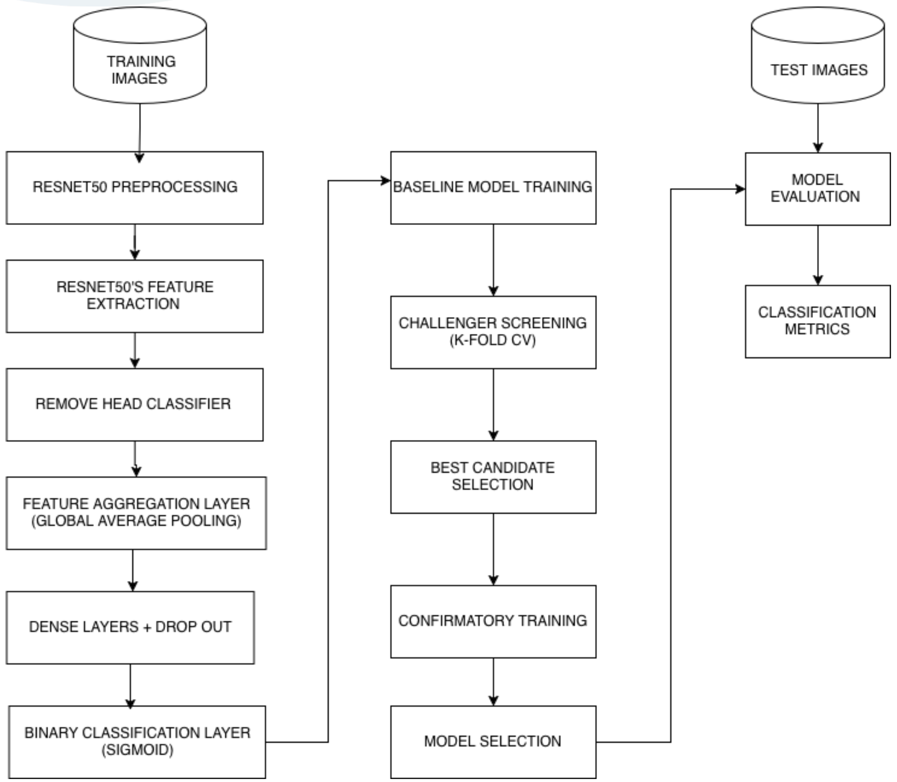
Support Vector Classifier (SVM) using RBF kernel (Ruzavina et al., 2025)
- Hyperparameters
  - C
    - controls error tolerance
  - Gamma
    - controls boundary shape

Stratified K-fold Cross-Validation on training data
- K=5

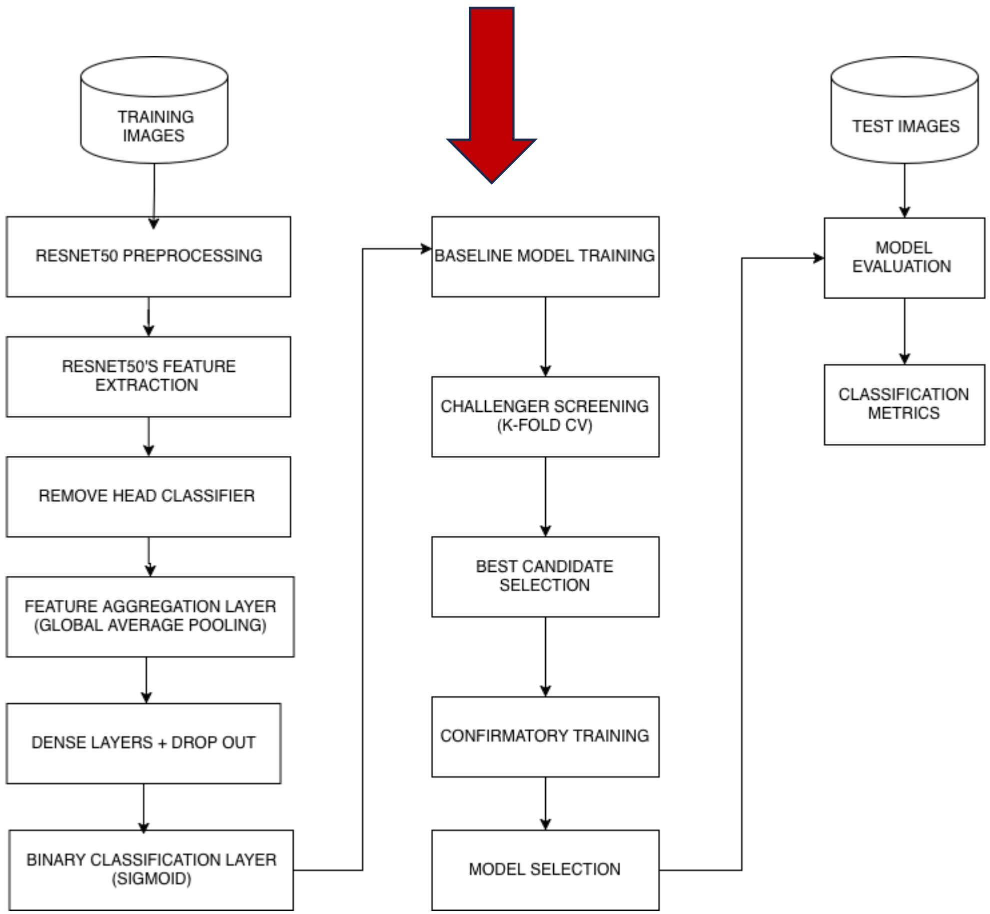
Best hyperparameters selection

Evaluate cross validated model on test data
Extract classification metrics

Deep Learning ResNet50 Pipeline



- TensorFlow + Keras
  - High-level API, readable code
- Use a pre-trained Convolutional Neural Network CNN (ResNet50) to enhance classification (Ruzavina et al., 2025) .
- Steps:
  - Resize images to 224x224
  - RGB images aim to leverage color relationships
  - Remove ResNet50 head classifier
  - Add aggregation layer (Global Average Pooling)
  - Add Dense Layers
  - Add Dropout Regularization to reduce overfitting
    - (Srivastava et al., 2014)
  - Add Binary Classification Layer
- Hyperparameters
  - Learning rate
  - Dense units
  - Dropout rates



- Define baseline and 3 challengers hyperparameters sets
  - C1,C2,C3
- Baseline model training
- Run quick K-fold CV on C1,C2,C3
  - K=2
  - Epochs=2
- Select best challenger
- Run confirmatory training on selected challenger
- Select challenger if F1 improves respect to baseline's

COMPARISON OUTCOMES

CLASSICAL VS DEEP LEARNING

Performance

Metric	ResNet50	SVM	Better
Accuracy	0.9320	0.9360	SVM
Weighted F1	0.9313	0.9361	SVM
Macro F1	0.9301	0.9355	SVM
Not-good Recall	0.8571	0.9464	SVM
Good Recall	0.9928	0.9275	ResNet50

Resnet50 CM		
Actual / Predicted	Good	Not-good
Good	137	1
Not-good	16	96

SVM CM		
Actual / Predicted	Good	Not-good
Good	128	10
Not-good	6	106

SVM is better overall

Bias between classes

Metric	ResNet50	SVM
Accuracy	0.9320	0.9360
Weighted F1	0.9313	0.9361
Macro F1	0.9301	0.9355
Not-good Recall	0.8571	0.9464
Good Recall	0.9928	0.9275
Recall gap	**+0.1357**	**-0.0189**

- ResNet50 biased towards good class
  - Tends to minimize false alarms
- SVM slight biased towards not-good class
  - Detects more defects

Computational Cost

Stage	ResNet50 Time *	SVM Time	Better
Preprocessing	0.244 s	106.594 s	
Training	554.146 s	137.717 s	
Total before Evaluation	**554.390 s**	**244.311 s**	**SVM**
Evaluation (250 samples)	**8.133 s**	**0.0173 (inference only) + 5.3809 (feature extraction) = 5.398 s**	**SVM**

* ResNet50 needed to use GPU, whereas SVM does not

Testing Environment: MacBook M4 equipped with 16 GB

Cyclomatic Complexity *

Stage	SVM	ResNet50	Better
Preprocessing	6	10	SVM
Modelling & Validation	5	7	SVM
Evaluation	2	2	
Total	**13**	**19**	**SVM**

* Calculated via Python Abstract Syntax Tree (AST)

Lighting Robustness

- Check whether predictions stay stable under controlled illumination perturbations.
- Perturbations include darkening, brightening, additive offsets, and contrast shifts.
- After perturbations, extract performance metrics and measure the change
- Criteria
  - **accuracy >=90%**
  - **flip_rate <= 7%**
  - **Good/not-good recall delta <= 2%**
- Claim
  - Model must be robust on moderate lighting perturbations

Lighting Robustness strategy

- SVM(GLCM+DWT)
    - Percentile Normalization:
    - Global scaling method that remaps pixels.
    - Reduces the influence of extreme outliers.
    - 2nd–98th percentiles were selected
- Transfer Learning
    - CLAHE (Contrast Limited Adaptive Histogram Equalization)
    - Local contrast enhancement technique.
    - Divides images into tiles to perform histogram equalization
    - Clip-limit parameter was set to 0.003
- Key finding
    - Global normalization works better with handcrafted features (GLCM/DWT)
    - Local contrast enhancement suits automatically learned features (Transfer learning)

Lighting Robustness

	SVM					ResNet				
Perturbation	Accuracy	Flips	Flip Rate	Good Recall	Not-Good Recall	Accuracy	Flips	Flip Rate	Good Recall	Not-Good Recall
Baseline	0.936	0	0.000	0.928	0.946	0.928	0	0.000	1.000	0.839
Dark (-25%)	0.936	0	0.000	0.928	0.946	0.928	0	0.000	1.000	0.839
Dark (-50%)	0.936	0	0.000	0.928	0.946	0.928	0	0.000	1.000	0.839
Offset (+10)	0.936	0	0.000	0.928	0.946	0.916	3	0.012	0.993	0.821
Offset (-10)	0.932	3	0.012	0.920	0.946	0.928	4	0.016	0.993	0.848
Low Contrast	0.936	0	0.000	0.928	0.946	0.928	0	0.000	1.000	0.839
High Contrast	0.932	3	0.012	0.920	0.946	0.920	6	0.024	0.986	0.839
Bright (+25%)	0.940	5	0.020	0.942	0.938	0.916	5	0.020	0.986	0.830
Bright (+50%)	0.792	50	0.200	0.681	0.929	0.884	17	0.068	0.942	0.812

- Moderate perturbations:
    - Both passed Robustness Criteria(accuracy >=90%, flip_rate <= 7%, Good/not-good recall delta <= 2%)
    - SVM is better
- Severe overexposure:
    - ResNet is better

Occlusion Sensitivity (Explainable AI)

- Occlusion sensitivity heatmaps
    - Model-agnostic, perturbation-based XAI method (van der Velden et al., 2022)
    - It masks image regions and measures the change in prediction confidence, generating a heatmap of influential regions.
    - Applying it to both models allows assessing whether explanations from each model are consistent or divergent.

Occlusion Sensitivity

Case (samples = 5)	SVM Mean Confidence	SVM Mean Max Activation	ResNet50 Mean Confidence	ResNet50 Max Activation
True Positive (TP)	0.9930	0.00663	1.0000	0.00000
False Positive (FP)	0.8172	0.05318	0.6525	0.00000
False Negative (FN)	0.7841	0.07051	0.99936	0.81508
True Negative (TN)	0.9904	0.01831	1.0000	0.06801

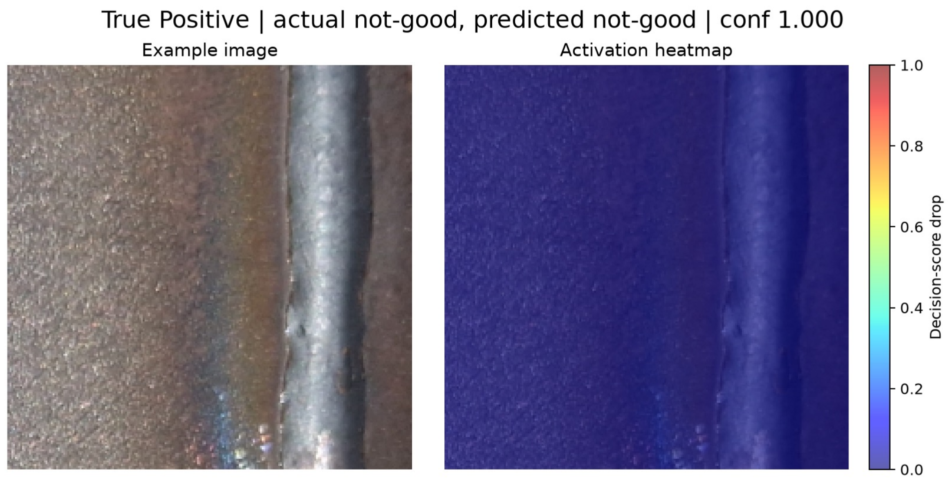
- Confidence:
  - SVM is informative
  - ResNet50’ overconfidence make it less useful
- Activation
  - SVM distributes attention evenly
  - ResNet50 overfocuses on specific regions

Occlusion Sensitivity

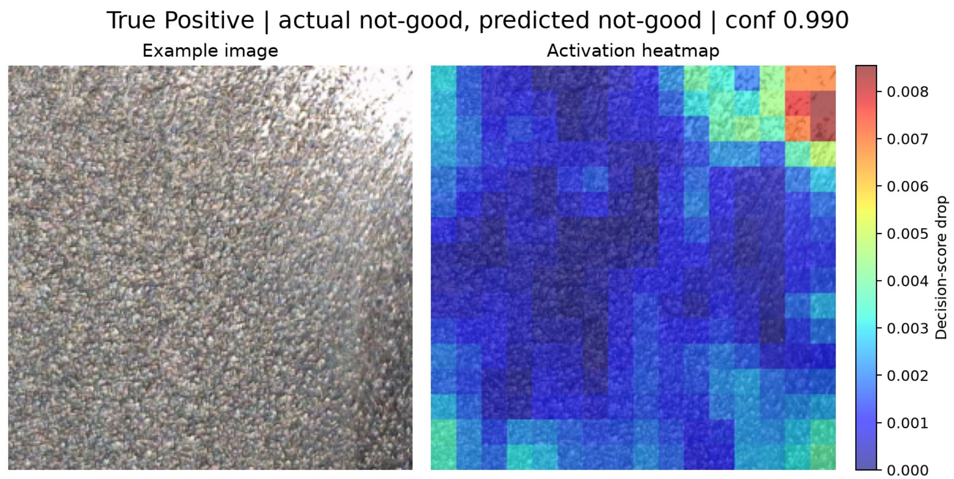
- True positives
    - Representative samples

- Overall assessment
    - SVM
        - Confidence is better calibrated
        - Heatmaps are more interpretable
    - Resnet50
        - Highly confident even on errors
        - Heatmaps less informative

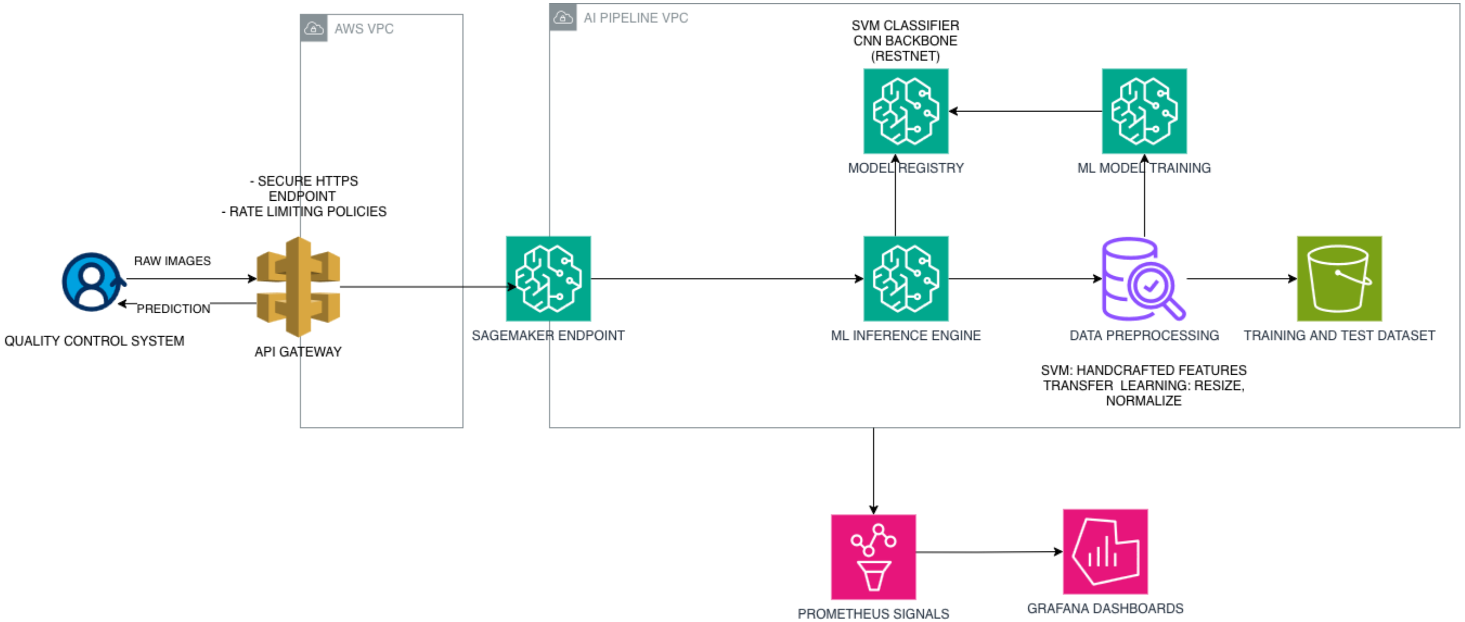
ResNet50



SVM



Deployment



- ResNet50 benefits more from this AWS SageMaker
- GPU, inference endpoint
- For Simpler SVM, AWS ECS + FastAPI would be enough

Ethics

- Algorithmic fairness in Quality Control
  - SVM is better at detecting defective parts
    - Favors customer protection
  - ResNet50 is less likely to classify good parts as defective
    - Favors production efficiency
- Real-world consequences
  - Defective part is accepted
    - Product failures, Warranty claims -> Reputational damage
  - Good part flagged as defective
    - Rework, manual inspection -> Reduced throughput
- Environmental Bias (Lighting)
  - Risk: Model dependency on shifts or camera settings.
- XAI Bias
  - Confidence
    - ResNet50 Risk: Overconfident scores bypassing manual reviews.
  - Activation
    - SVM Risk: Overlooking subtle defects

Overall ethical recommendation for production deployment: SVM

- Fairer towards customer safety
- Better overall performance metrics
- Lower compute burden and complexity
- More robust to lighting perturbations
- Its explanations are more useful and calibrated

PROJECT DEMONSTRATION

The complete code, documentation, and comparative evaluation results are available at

https://github.com/luistorresz/steelblast-machine-learning-project

REFERENCES

- Ruzavina, I., Theis, L.S., Lemeer, J., de Groen, R., *et al.* (2025) 'SteelBlastQC Dataset'. DataverseNL. Available at: https://doi.org/10.34894/EKZNN0.
- Ruzavina, I., Theis, L.S., Lemeer, J., Groen, R. de, *et al.* (2025) 'SteelBlastQC: Shot-blasted Steel Surface Dataset with Interpretable Detection of Surface Defects'. arXiv. Available at: https://doi.org/10.48550/arXiv.2504.20510.
- Srivastava, N. *et al.* (2014) 'Dropout: A Simple Way to Prevent Neural Networks from Overfitting', *Journal of Machine Learning Research* [Preprint].
- Stachowiak, G.P., Podsiadlo, P. and Stachowiak, G.W. (2005) 'A Comparison of Texture Feature Extraction Methods for Machine Condition Monitoring and Failure Analysis', *Tribology Letters*, 20(2), pp. 133–147. Available at: https://doi.org/10.1007/s11249-005-8303-1.
- van der Velden, B.H.M. et al. (2022) 'Explainable artificial intelligence (XAI) in deep learning-based medical image analysis', Medical Image Analysis, 79, p. 102470. Available at: https://doi.org/10.1016/j.media.2022.102470.